Electricity & Magnetism

Option B: Important Derivations Register

Exam Revision Version

May 31, 2026

Contents

1 Derivation 1: Gauss's Theorem (Most Important)

PYQ Weightage

PYQ Weightage: ⋆⋆⋆⋆⋆

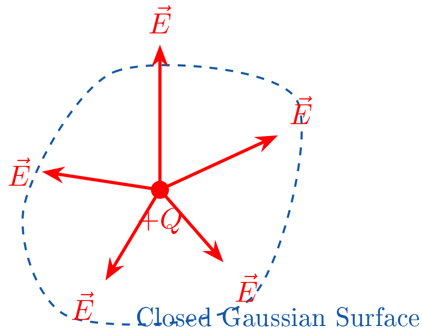
Question

State and prove Gauss's theorem in electrostatics.

Statement

The total electric flux through any closed surface is equal to $\frac{1}{\epsilon_0}$ times the total charge enclosed within the surface.

Diagram



Proof

Consider a point charge Q enclosed inside a closed surface. Electric field at a point on the surface:

$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

Small area element:

$$dA$$

Electric flux through area element:

$$d\phi = \vec{E} \cdot d\vec{A}$$

$$d\phi = E\,dA\cos\theta$$

But

$$d\Omega = \frac{dA\cos\theta}{r^2}$$

where $d\Omega$ is the solid angle.

Hence

$$d\phi = \frac{Q}{4\pi\epsilon_0}d\Omega$$

Integrating over complete closed surface,

$$\phi = \frac{Q}{4\pi\epsilon_0}\int d\Omega$$

Total solid angle around a point:

$$\int d\Omega = 4\pi$$

Therefore

$$\phi = \frac{Q}{4\pi\epsilon_0}(4\pi)$$

$$\boxed{\phi = \frac{Q}{\epsilon_0}}$$

Final Result

$$\boxed{\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}}$$

Differential Form

Using Divergence Theorem,

$$\boxed{\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}}$$

Exam Ending Line

Hence proved that the total electric flux through any closed surface equals the enclosed charge divided by ϵ_0.

2 Derivation 2: Electric Field Due to Infinite Line Charge Using Gauss Law

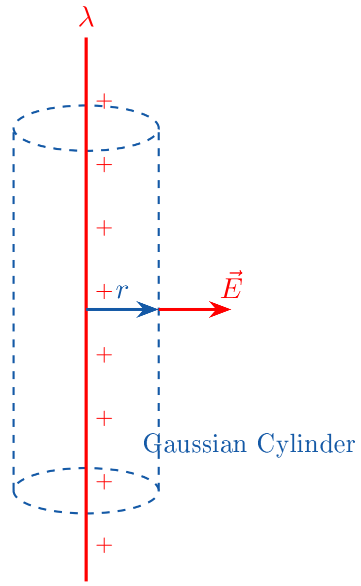
PYQ Weightage

PYQ Weightage: ∗∗∗∗∗

Question

Using Gauss's theorem, obtain the electric field due to an infinitely long straight line charge.

Diagram



Given

- Linear charge density: λ
- Radius of Gaussian cylinder: r
- Length: l

Step 1

Choose cylindrical Gaussian surface. By symmetry, the electric field is radial and constant over the curved surface.

Step 2: Apply Gauss Law

$$\oint E\,dA = \frac{Q}{\epsilon_0}$$

Step 3

Flux through end caps $= 0$ because $\vec{E} \perp d\vec{A}$. Only curved surface contributes. Curved surface area:

$$A = 2\pi r l$$

Thus

$$E(2\pi rl) = \frac{\lambda l}{\epsilon_0}$$

Step 4

Cancel l:

$$E(2\pi r) = \frac{\lambda}{\epsilon_0}$$

Hence

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Final Result

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Important Note

$$E \propto \frac{1}{r}$$

Exam Ending Line

Hence the electric field due to an infinite line charge decreases inversely with distance.

3 Derivation 3: Electric Potential Due to an Electric Dipole

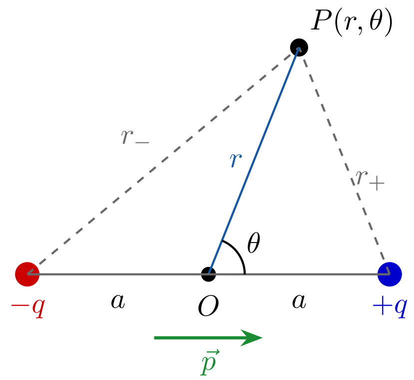
PYQ Weightage

PYQ Weightage: ∗∗∗∗∗ Direct PYQ (Set 1 & Set 3)

Question

Obtain an expression for the electric potential due to an electric dipole at a point $P(r, \theta)$.

Diagram



Given

- Dipole charges: $-q, \quad +q$
- Separation: $2a$
- Dipole moment:

$$\boxed{p = 2aq}$$

Step 1: Potential Due to Individual Charges

Potential due to positive charge:

$$V_+ = \frac{1}{4\pi\epsilon_0}\frac{q}{r_+}$$

Potential due to negative charge:

$$V_- = -\frac{1}{4\pi\epsilon_0}\frac{q}{r_-}$$

Step 2: Total Potential

By superposition,

$$V = V_+ + V_-$$

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r_+} - \frac{q}{r_-}\right)$$

Step 3: For a Distant Point

When $r \gg a$, we use

$$r_+ = r - a\cos\theta$$

$$r_- = r + a\cos\theta$$

Substituting,

$$V = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{r - a\cos\theta} - \frac{1}{r + a\cos\theta}\right]$$

Step 4: Simplification

Using

$$\frac{1}{x-y} - \frac{1}{x+y} = \frac{2y}{x^2 - y^2}$$

we obtain

$$V = \frac{q}{4\pi\epsilon_0}\frac{2a\cos\theta}{r^2}$$

Since $p = 2aq$, therefore

$$\boxed{V = \frac{1}{4\pi\epsilon_0}\frac{p\cos\theta}{r^2}}$$

Final Result

$$V = \frac{1}{4\pi\epsilon_0}\frac{p\cos\theta}{r^2}$$

Special Cases

Axial Position ($\theta = 0°$)

$$V = \frac{1}{4\pi\epsilon_0}\frac{p}{r^2}$$

Equatorial Position ($\theta = 90°$)

$$V = 0$$

Exam Ending Line

Hence the electric potential due to a short dipole at a distant point is

$$V = \frac{1}{4\pi\epsilon_0}\frac{p\cos\theta}{r^2}$$

4 Derivation 4: Electric Field Due to an Electric Dipole

PYQ Weightage

PYQ Weightage: *** Direct PYQ (Set 1 & Set 3)**

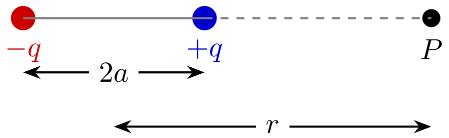
Question

Obtain expressions for electric field due to an electric dipole on:

1. Axial position
2. Equatorial position

(A) Electric Field on Axial Line

Diagram



Step 1

Field at P due to $+q$:

$$E_+ = \frac{1}{4\pi\epsilon_0}\frac{q}{(r-a)^2}$$

Direction: Right

Step 2

Field due to $-q$:

$$E_- = \frac{1}{4\pi\epsilon_0}\frac{q}{(r+a)^2}$$

Direction: Left

Step 3

Net Field:

$$E = E_+ - E_-$$

$$E = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2}\right]$$

Step 4

Simplifying,

$$E = \frac{1}{4\pi\epsilon_0}\frac{4aqr}{(r^2-a^2)^2}$$

For $r \gg a$,

$$r^2 - a^2 \approx r^2$$

Therefore

$$E = \frac{1}{4\pi\epsilon_0}\frac{4aq}{r^3}$$

Since $p = 2aq$,

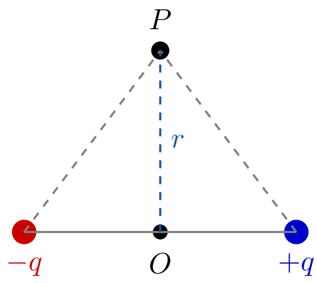
$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0}\frac{2p}{r^3}$$

Final Result (Axial)

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0}\frac{2p}{r^3}$$

(B) Electric Field on Equatorial Line

Diagram



Step 1

Distance of P from each charge:

$$r = \sqrt{x^2 + a^2}$$

Field due to each charge:

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$$

Step 2

Resolve into components. Vertical components cancel. Horizontal components add.

Step 3

Net Field after simplification:

$$E = \frac{1}{4\pi\epsilon_0}\frac{2qa}{r^3}$$

Using $p = 2aq$,

$$E_{\text{equatorial}} = \frac{1}{4\pi\epsilon_0}\frac{p}{r^3}$$

Final Result (Equatorial)

$$E_{\text{equatorial}} = \frac{1}{4\pi\epsilon_0}\frac{p}{r^3}$$

Comparison

Table 1: Electric Field Comparison

Position	Electric Field
Axial	$\frac{1}{4\pi\epsilon_0}\frac{2p}{r^3}$
Equatorial	$\frac{1}{4\pi\epsilon_0}\frac{p}{r^3}$

5 Derivation 5: Biot–Savart Law

PYQ Weightage

PYQ Weightage: *** Direct PYQ (Set 1, Set 2, Set 3)**

Question

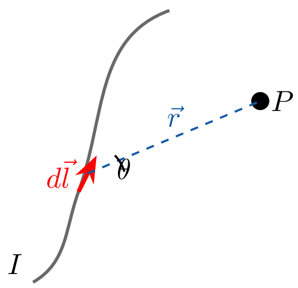
State and derive Biot–Savart Law.

Statement

The magnetic field at a point due to a small current element is:

1. Directly proportional to current (I)
2. Directly proportional to current element (dl)
3. Directly proportional to $\sin\theta$
4. Inversely proportional to square of distance (r^2)

Diagram



Experimental Observations

Magnetic field (dB) is:

- Directly proportional to current:
$$dB \propto I$$
- Directly proportional to current element:
$$dB \propto dl$$

- Directly proportional to $\sin\theta$:

$$dB \propto \sin\theta$$

- Inversely proportional to square of distance:

$$dB \propto \frac{1}{r^2}$$

Combining all:

$$dB \propto \frac{I\,dl\sin\theta}{r^2}$$

Introducing proportionality constant $\frac{\mu_0}{4\pi}$, hence

$$dB = \frac{\mu_0}{4\pi}\frac{I\,dl\sin\theta}{r^2}$$

Vector Form

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I(d\vec{l}\times\hat{r})}{r^2}$$

Special Cases

- **Case 1:** $\theta = 0^\circ$

$$\sin 0^\circ = 0$$

$$dB = 0$$

- **Case 2:** $\theta = 90^\circ$

$$\sin 90^\circ = 1$$

$$dB = \frac{\mu_0}{4\pi}\frac{I\,dl}{r^2}$$

Maximum magnetic field.

Final Result

$$dB = \frac{\mu_0}{4\pi}\frac{I\,dl\sin\theta}{r^2}$$

Exam Ending Line

Hence proved the Biot–Savart law for magnetic field due to a current element.

6 Derivation 6: Magnetic Field Due to Infinite Straight Wire

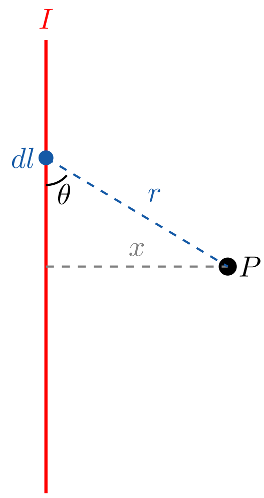
PYQ Weightage

PYQ Weightage: *** Direct PYQ**

Question

Using Biot–Savart law, obtain an expression for magnetic field due to an infinitely long straight current-carrying conductor.

Diagram



Step 1

From Biot–Savart Law,

$$dB = \frac{\mu_0}{4\pi}\frac{I\,dl\sin\theta}{r^2}$$

Step 2

From geometry,

$$r = x\csc\theta$$

and

$$dl = x\csc^2\theta\,d\theta$$

Step 3

Substitute in Biot–Savart law:

$$dB = \frac{\mu_0 I}{4\pi}\frac{x\csc^2\theta\sin\theta\,d\theta}{x^2\csc^2\theta}$$

Simplifying,

$$dB = \frac{\mu_0 I}{4\pi x}\sin\theta\,d\theta$$

Step 4

Integrate for infinite wire: $\theta = -90^\circ$ to $\theta = +90^\circ$. Thus

$$B = \frac{\mu_0 I}{4\pi x} \int_{-\pi/2}^{+\pi/2} \sin\theta \, d\theta$$

Step 5

$$B = \frac{\mu_0 I}{4\pi x} \left[-\cos\theta\right]_{-\pi/2}^{+\pi/2}$$

$$B = \frac{\mu_0 I}{4\pi x}(2)$$

Therefore

$$\boxed{B = \frac{\mu_0 I}{2\pi x}}$$

Final Result

$$\boxed{B = \frac{\mu_0 I}{2\pi r}}$$

Exam Ending Line

Hence the magnetic field due to an infinitely long straight conductor varies inversely with distance from the conductor.

7 Derivation 7: Magnetic Field on Axis of Circular Coil

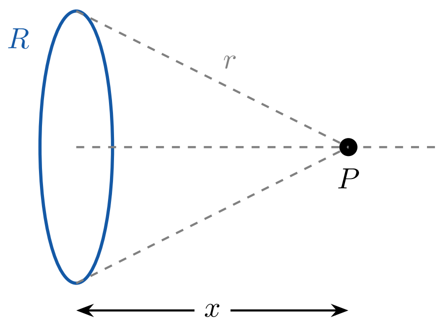
PYQ Weightage

PYQ Weightage: ∗∗∗∗∗ Direct 10-Mark PYQ (Set 1)

Question

Derive an expression for magnetic field at a point on the axis of a circular current-carrying coil.

Diagram



Step 1

Consider a small current element (dl). Using Biot–Savart Law,

$$dB = \frac{\mu_0}{4\pi}\frac{I\,dl}{r^2}$$

because $\theta = 90^\circ$.

Step 2

Distance of point (P) from every element:

$$r = \sqrt{R^2 + x^2}$$

Step 3

Resolve (dB) into components. Due to symmetry, perpendicular components cancel while axial components add. Hence

$$dB_x = dB\cos\phi$$

Step 4

Substitute:

$$dB_x = \frac{\mu_0}{4\pi}\frac{I\,dl}{r^2}\cos\phi$$

From geometry,

$$\cos\phi = \frac{R}{r}$$

Therefore

$$dB_x = \frac{\mu_0}{4\pi}\frac{IR\,dl}{r^3}$$

Step 5

Integrate around complete coil:

$$B = \frac{\mu_0 I R}{4\pi r^3} \oint dl$$

But

$$\oint dl = 2\pi R$$

Hence

$$B = \frac{\mu_0 I R}{4\pi r^3}(2\pi R)$$

$$B = \frac{\mu_0 I R^2}{2r^3}$$

Substitute $r = \sqrt{R^2 + x^2}$.

Final Result

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$

For N Turns Coil

$$B = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}}$$

At Centre ($x = 0$)

$$B = \frac{\mu_0 N I}{2R}$$

8 Derivation 8: Ampere's Circuital Law

PYQ Weightage

PYQ Weightage: *** Important Derivation**

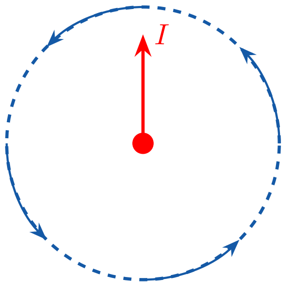
Question

State and explain Ampere's Circuital Law.

Statement

The line integral of magnetic field around any closed path is equal to μ_0 times the net current enclosed by the path.

Diagram



Amperian Loop

Mathematical Form

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

Explanation

Consider a current carrying conductor enclosed by a circular path. Magnetic field at every point on the circular path is constant. Also

$$\vec{B} \parallel d\vec{l}$$

Therefore

$$\vec{B} \cdot d\vec{l} = B\,dl$$

Hence

$$\oint \vec{B} \cdot d\vec{l} = B \oint dl$$

Since total enclosed current is I,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

Final Result

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

Exam Ending Line

Hence proved Ampere's Circuital Law.

9 Derivation 9: Magnetic Field Inside a Long Solenoid

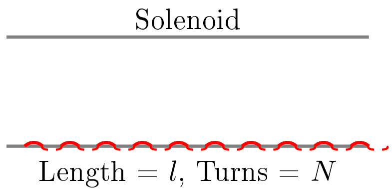
PYQ Weightage

PYQ Weightage: *** Direct PYQ**

Question

Using Ampere's Circuital Law, obtain an expression for magnetic field inside a long solenoid.

Diagram



Step 1

Choose rectangular Amperian loop.

Step 2: Apply Ampere's Law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

Step 3

Outside solenoid:

$$B \approx 0$$

Hence only inside path contributes.

$$Bl = \mu_0 I_{\text{enc}}$$

Step 4

Turns per unit length:

$$n = \frac{N}{l}$$

Number of turns enclosed: nl. Current enclosed:

$$I_{\text{enc}} = nlI$$

Step 5

Substitute:

$$Bl = \mu_0(nlI)$$

Cancel l:

$$\boxed{B = \mu_0 nI}$$

Final Result

$$\boxed{B = \mu_0 nI}$$

Solenoid With Core

$$\boxed{B = \mu_0 \mu_r nI}$$

Exam Ending Line

Hence magnetic field inside a long solenoid is uniform and equals

$$\boxed{B = \mu_0 nI}$$

10 Derivation 10: Relation Between Permeability and Susceptibility

PYQ Weightage

PYQ Weightage: ** Direct PYQ**

Question

Prove that

$$\mu = \mu_0(1 + \chi)$$

Step 1

Magnetic induction in a material:

$$B = \mu_0(H + M)$$

Step 2

Using $M = \chi H$, substitute:

$$B = \mu_0(H + \chi H)$$

$$B = \mu_0 H(1 + \chi)$$

Step 3

But $B = \mu H$, therefore

$$\mu H = \mu_0 H(1 + \chi)$$

Cancel H:

$$\mu = \mu_0(1 + \chi)$$

Final Result

$$\mu = \mu_0(1 + \chi)$$

Also

$$\mu_r = 1 + \chi$$

11 Derivation 11: Faraday's Second Law

PYQ Weightage

PYQ Weightage: *** Direct PYQ**

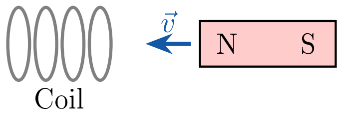
Question

State and derive Faraday's Second Law of Electromagnetic Induction.

Statement

The induced emf is equal to the negative rate of change of magnetic flux linked with the circuit.

Diagram



Step 1

Let magnetic flux change from ϕ_1 to ϕ_2 in time Δt.

Step 2

Experimentally,

$$e \propto \frac{\Delta\phi}{\Delta t}$$

Step 3

Introducing negative sign according to Lenz's Law,

$$e = -\frac{\Delta\phi}{\Delta t}$$

For infinitesimal changes,

$$e = -\frac{d\phi}{dt}$$

For N Turns Coil

$$e = -N\frac{d\phi}{dt}$$

Final Result

$$e = -N\frac{d\phi}{dt}$$

12 Derivation 12: Self-Inductance of a Long Solenoid

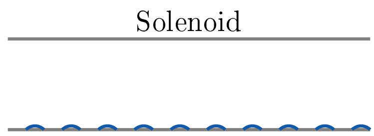
PYQ Weightage

PYQ Weightage: ∗∗∗∗∗ Direct PYQ

Question

Derive an expression for self-inductance of a long solenoid.

Diagram



Length $= l$, Area $= A$, Turns $= N$

Step 1

Magnetic field inside solenoid:

$$B = \mu_0 n I$$

where $n = \frac{N}{l}$. Hence

$$B = \mu_0 \frac{N}{l} I$$

Step 2

Magnetic flux through one turn:

$$\phi = BA$$

Substituting,

$$\phi = \mu_0 \frac{N}{l} IA$$

Step 3

Flux linkage:

$$N\phi = N\left(\mu_0 \frac{N}{l} IA\right)$$

$$N\phi = \mu_0 \frac{N^2 A}{l} I$$

Step 4

Using $L = \frac{N\phi}{I}$, therefore

$$\boxed{L = \frac{\mu_0 N^2 A}{l}}$$

Final Result

$$L = \frac{\mu_0 N^2 A}{l}$$

13 Derivation 13: Continuity Equation

PYQ Weightage

PYQ Weightage: ∗∗∗∗∗ Direct PYQ (Set 2)

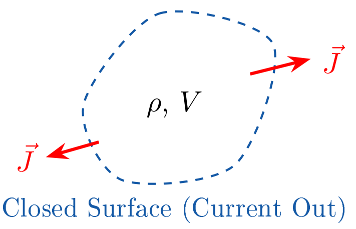
Question

Derive the Continuity Equation and explain its physical significance.

Statement

The continuity equation expresses the law of conservation of electric charge.

Diagram



Step 1: Charge Inside Volume

Let charge density inside volume V be ρ. Total charge inside volume:

$$Q = \int \rho \, dV$$

Step 2: Current Leaving Surface

Current density: $\vec{J}$. Total current flowing out:

$$I = \oint \vec{J} \cdot d\vec{S}$$

Step 3: Conservation of Charge

Rate of decrease of charge inside volume equals current flowing out.

$$\oint \vec{J} \cdot d\vec{S} = -\frac{d}{dt} \int \rho \, dV$$

Step 4: Apply Divergence Theorem

$$\oint \vec{J} \cdot d\vec{S} = \int (\nabla \cdot \vec{J})\, dV$$

Substituting,

$$\int (\nabla \cdot \vec{J})\, dV = -\int \frac{\partial \rho}{\partial t}\, dV$$

Step 5: Equate Integrands

Since equation is valid for every volume,

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

Therefore

$$\boxed{\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0}$$

Final Result

$$\boxed{\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0}$$

Physical Significance

Represents:

Conservation of Charge

Charge can neither be created nor destroyed.

Exam Ending Line

Hence proved the continuity equation.

14 Derivation 14: Maxwell's Equations

PYQ Weightage

PYQ Weightage: *** Direct 10-Mark PYQ**

Question

Write Maxwell's equations in integral and differential forms and explain their significance.

Maxwell's First Equation: Gauss's Law for Electricity

Integral Form

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$$

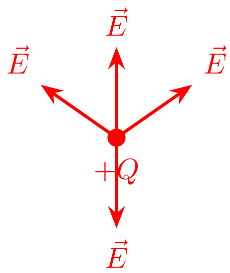
Differential Form

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Physical Meaning

Electric charges are sources of electric field.

Diagram



Maxwell's Second Equation: Gauss's Law for Magnetism

Integral Form

$$\oint \vec{B} \cdot d\vec{S} = 0$$

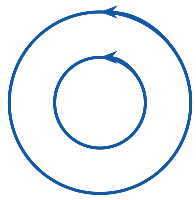
Differential Form

$$\nabla \cdot \vec{B} = 0$$

Physical Meaning

Magnetic monopoles do not exist.

Diagram



Magnetic Field Loops

Maxwell's Third Equation: Faraday's Law

Integral Form

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$$

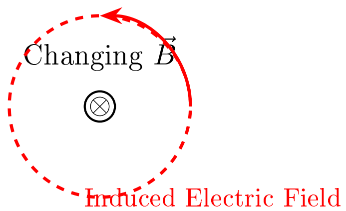
Differential Form

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Physical Meaning

Changing magnetic field produces electric field.

Diagram



Maxwell's Fourth Equation: Ampere-Maxwell Law

Integral Form

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

Differential Form

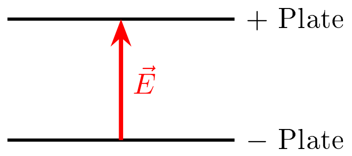
$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Physical Meaning

Magnetic field is produced by:

1. Conduction current
2. Changing electric field

Diagram



Maxwell Equation Summary Table

Table 2: Maxwell's Equations Summary

Equation	**Differential Form**
Gauss (Electric)	$\nabla \cdot \vec{E} = \rho/\epsilon_0$
Gauss (Magnetic)	$\nabla \cdot \vec{B} = 0$
Faraday Law	$\nabla \times \vec{E} = -\partial \vec{B}/\partial t$
Ampere-Maxwell Law	$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \partial \vec{E}/\partial t$

Exam Ending Line

Hence Maxwell unified electricity and magnetism into one theory.

15 Derivation 15: Thevenin's Theorem

PYQ Weightage

PYQ Weightage: *** Direct PYQ**

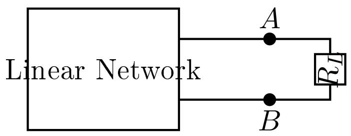
Question

State and explain Thevenin's theorem.

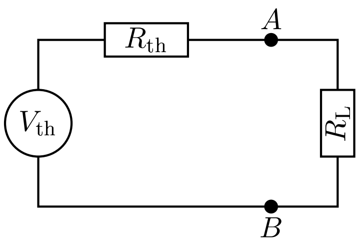
Statement

Any linear bilateral network can be replaced by an equivalent voltage source (V_{th}) in series with an equivalent resistance (R_{th}).

Original Circuit



Equivalent Circuit



Procedure

Step 1

Remove load resistor (R_L).

Step 2: Find open-circuit voltage

$$V_{\text{th}} = V_{\text{OC}}$$

Step 3: Deactivate sources

- Voltage source → Short circuit
- Current source → Open circuit

Step 4: Find equivalent resistance

$$R_{\text{th}}$$

Step 5: Reconnect load

Load Current

$$I = \frac{V_{\text{th}}}{R_{\text{th}} + R_L}$$

Final Result

Equivalent circuit becomes

$$V_{\text{th}} \text{ in series with } R_{\text{th}}$$

16 Derivation 16: Maximum Power Transfer Theorem

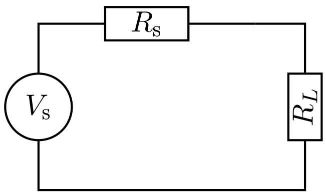
PYQ Weightage

PYQ Weightage: *** Direct PYQ**

Question

Prove that maximum power is transferred when load resistance equals source resistance.

Circuit



Step 1: Current

$$I = \frac{V_s}{R_s + R_L}$$

Step 2: Load Power

$$P = I^2 R_L$$

Substituting current,

$$P = \frac{V_s^2 R_L}{(R_s + R_L)^2}$$

Step 3: Differentiate

For maximum power,

$$\frac{dP}{dR_L} = 0$$

Differentiating,

$$(R_s + R_L) - 2R_L = 0$$
$$R_s - R_L = 0$$

Therefore

$$R_L = R_s$$

Condition for Maximum Power

$$R_L = R_s$$

or

$$R_L = R_{\text{th}}$$

Step 4: Maximum Power

Substitute $R_L = R_s$:

$$P_{\max} = \frac{V_s^2}{4R_s}$$

Final Result

$$\boxed{P_{\max} = \frac{V_s^2}{4R_s}}$$

Derivation Register Completed

Most Important Exam Derivations:

- ✓ Gauss's Theorem
- ✓ Electric Field due to Infinite Line Charge
- ✓ Potential due to Dipole
- ✓ Electric Field due to Dipole
- ✓ Biot–Savart Law
- ✓ Magnetic Field due to Infinite Straight Wire
- ✓ Magnetic Field on Axis of Circular Coil
- ✓ Ampere's Circuital Law
- ✓ Magnetic Field Inside Solenoid
- ✓ Relation $\mu = \mu_0(1 + \chi)$
- ✓ Faraday's Second Law
- ✓ Self-Inductance of Solenoid
- ✓ Continuity Equation
- ✓ Maxwell's Equations
- ✓ Thevenin's Theorem
- ✓ Maximum Power Transfer Theorem

These are the derivations most strongly supported by your syllabus and repeated across Set 1, Set 2, and Set 3 PYQs.

17 Derivation 17: Mutual Inductance of Two Coaxial Solenoids

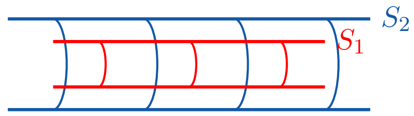
PYQ Weightage

PYQ Weightage: *** Important Syllabus Derivation**

Question

Derive an expression for the mutual inductance of two long coaxial solenoids.

Diagram



Length $= l$, Area $= A$, turns N_1, N_2

Given

Primary Solenoid:

- N_1 turns
- Current: I_1

Secondary Solenoid:

- N_2 turns

Common parameters:

- Common Area: A
- Length: l

Step 1: Magnetic Field Produced by Primary Solenoid

Magnetic field inside long solenoid:

$$B = \mu_0 n I$$

where $n = \frac{N_1}{l}$. Hence

$$B = \mu_0 \frac{N_1}{l} I_1$$

Step 2: Magnetic Flux Through One Turn of Secondary

$$\phi = BA$$

Substituting B,

$$\phi = \mu_0 \frac{N_1}{l} I_1 A$$

Step 3: Total Flux Linkage

Flux linkage with secondary coil:

$$N_2\phi = N_2\left(\mu_0\frac{N_1}{l}I_1A\right)$$

$$N_2\phi = \mu_0\frac{N_1N_2A}{l}I_1$$

Step 4: Definition of Mutual Inductance

$$\boxed{M = \frac{N_2\phi}{I_1}}$$

Substituting,

$$M = \frac{\mu_0\frac{N_1N_2A}{l}I_1}{I_1}$$

Cancelling I_1,

$$\boxed{M = \frac{\mu_0N_1N_2A}{l}}$$

Final Result

$$\boxed{M = \frac{\mu_0N_1N_2A}{l}}$$

Important Observations

$$M \propto N_1$$

$$M \propto N_2$$

$$M \propto A$$

$$M \propto \frac{1}{l}$$

Exam Ending Line

Hence the mutual inductance of two long coaxial solenoids is

$$\boxed{M = \frac{\mu_0N_1N_2A}{l}}$$

18 Derivation 18: Displacement Current and Modified Ampere's Law

PYQ Weightage

PYQ Weightage: ∗∗∗∗∗ Direct PYQ (Set 3)

Question

Explain displacement current and derive the modified Ampere-Maxwell equation.

Problem with Ampere's Law

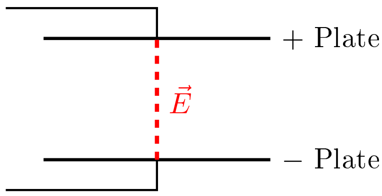
Ampere's law is

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

This works for steady currents.

Problem: Charging Capacitor

Diagram



Current exists in wires. But between capacitor plates:

$$I = 0$$

(no conduction current).

Yet magnetic field still exists. This creates a contradiction.

Maxwell's Idea

Maxwell proposed that a changing electric field behaves like a current. This is called:

Displacement Current

Electric Flux

Electric flux:

$$\phi_E = \int \vec{E} \cdot d\vec{S}$$

Displacement Current

Maxwell defined

$$I_d = \epsilon_0 \frac{d\phi_E}{dt}$$

Displacement Current Density

$$J_d = \epsilon_0 \frac{\partial E}{\partial t}$$

Modification of Ampere's Law

Original Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

Maxwell added displacement current term. Thus

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I + I_d)$$

Substituting

$$I_d = \epsilon_0 \frac{d\phi_E}{dt}$$

we get

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

Final Result

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

This is called the Ampere-Maxwell Equation.

Differential Form

Using Stokes' theorem,

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Physical Significance

Magnetic field can be produced by: Conduction Current I and Changing Electric Field $\frac{\partial E}{\partial t}$.

Exam Ending Line

Hence Maxwell removed the inconsistency in Ampere's law by introducing displacement current and obtained

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

Summary of Key Derivations

Register Complete ✓

This register contains the 18 high-yield derivations required for the E&M curriculum:

1. Gauss's Theorem
2. Electric Field due to Infinite Line Charge
3. Potential due to Dipole
4. Electric Field due to Dipole
5. Biot–Savart Law
6. Magnetic Field due to Infinite Straight Wire
7. Magnetic Field on Axis of Circular Coil
8. Ampere's Circuital Law
9. Magnetic Field Inside Solenoid
10. Relation $\mu = \mu_0(1 + \chi)$
11. Faraday's Second Law
12. Self-Inductance of Solenoid
13. Continuity Equation
14. Maxwell's Equations
15. Thevenin's Theorem
16. Maximum Power Transfer Theorem
17. Mutual Inductance of Two Coaxial Solenoids
18. Displacement Current & Ampere-Maxwell Law

All major analytical proofs across PYQ Sets 1, 2, and 3 are successfully detailed.